

Category Boundary Sorting Student Handout

TOK Cognitive Science Activity Bank • Knowledge and Language • 25-35 min

Category	Knowledge and Language	Time	25-35 min
Difficulty	Medium	ID	category-boundary-sorting

Big idea: Categories are made for purposes.

Student-Facing Prompt

Inspect Category Boundary Sorting Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Category Boundary Sorting Stimulus A	Student-facing stimulus 1: Cards such as game, sport, art, science, medicine, belief, evidence, and opinion. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Category Boundary Sorting Stimulus B	Student-facing stimulus 2: Borderline cases that force students to define category rules. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Use this for comparison, a second round, or a partner challenge.
Category Boundary Sorting Reveal / Twist	Student-facing stimulus 3: A rule revision round after students encounter exceptions. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Teacher reveal: connect the changed response to the big idea — Categories are made for purposes.

Actual Lesson Questions

#	Question for students
1	After inspecting Category Boundary Sorting Stimulus A, what is your first knowledge claim?
2	Which exact detail from Category Boundary Sorting Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	Do categories describe reality or organise it?

Ready-to-Use Examples

- Cards such as game, sport, art, science, medicine, belief, evidence, and opinion.
- Borderline cases that force students to define category rules.
- A rule revision round after students encounter exceptions.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Knowledge and Language.

Activity Steps

1. Groups sort prompts into yes, no and depends.
2. For each "depends", they must state the criterion.
3. Groups trade criteria and test them on borderline cases.
4. Discuss why different communities define the same term differently.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Do categories describe reality or organise it?
- Who has authority to define terms?
- When does disagreement about words become disagreement about reality?

Knowledge Questions

- Do categories describe reality or organise it?
- Who has authority to define terms?
- When does disagreement about words become disagreement about reality?

Transfer Example

For example, run this activity with cards such as game, sport, art, science, medicine, belief, evidence, and opinion. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...